K+ channel blocker

Azimilide

Normal cell line | **LQTS cell line** | **BrS cell line**

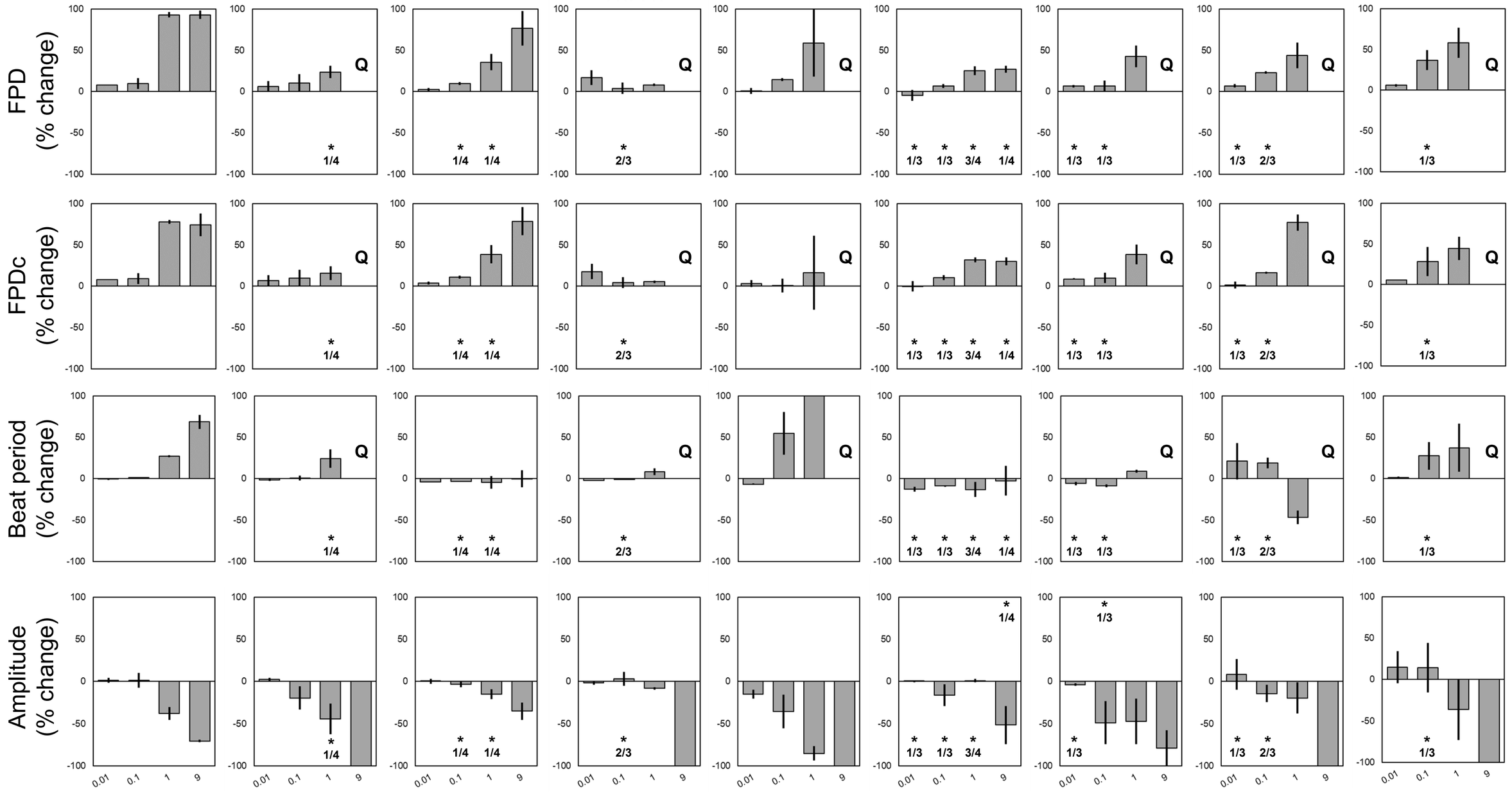
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



K+ channel blocker

Sotalol

Normal cell line | **LQTS cell line** | **BrS cell line**

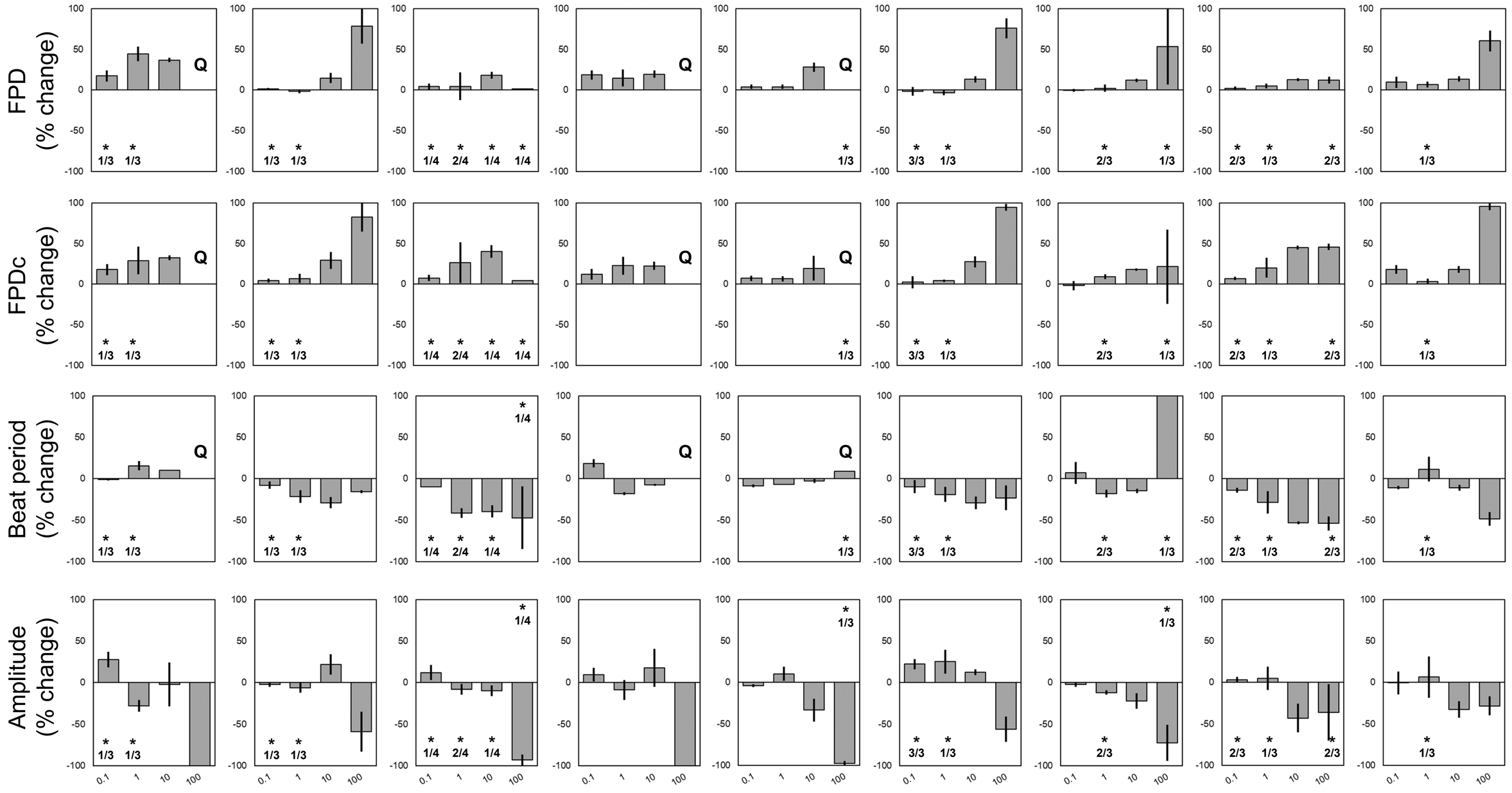
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Na⁺ channel blocker

Disopyramide

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

Q

* 1/4 | * 1/4 | * 1/4 | * 1/3 * 1/3 | | * 1/4 | | * 3/3 |

0.1 1 10 100

Na⁺ channel blocker

Mexiletine

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

Q

* 1/3 * 3/3 * 2/3 * 1/4

0.1 1 10 100

Ca2+ channel blocker

Bepridil

Normal cell line | **LQTS cell line** | **BrS cell line**

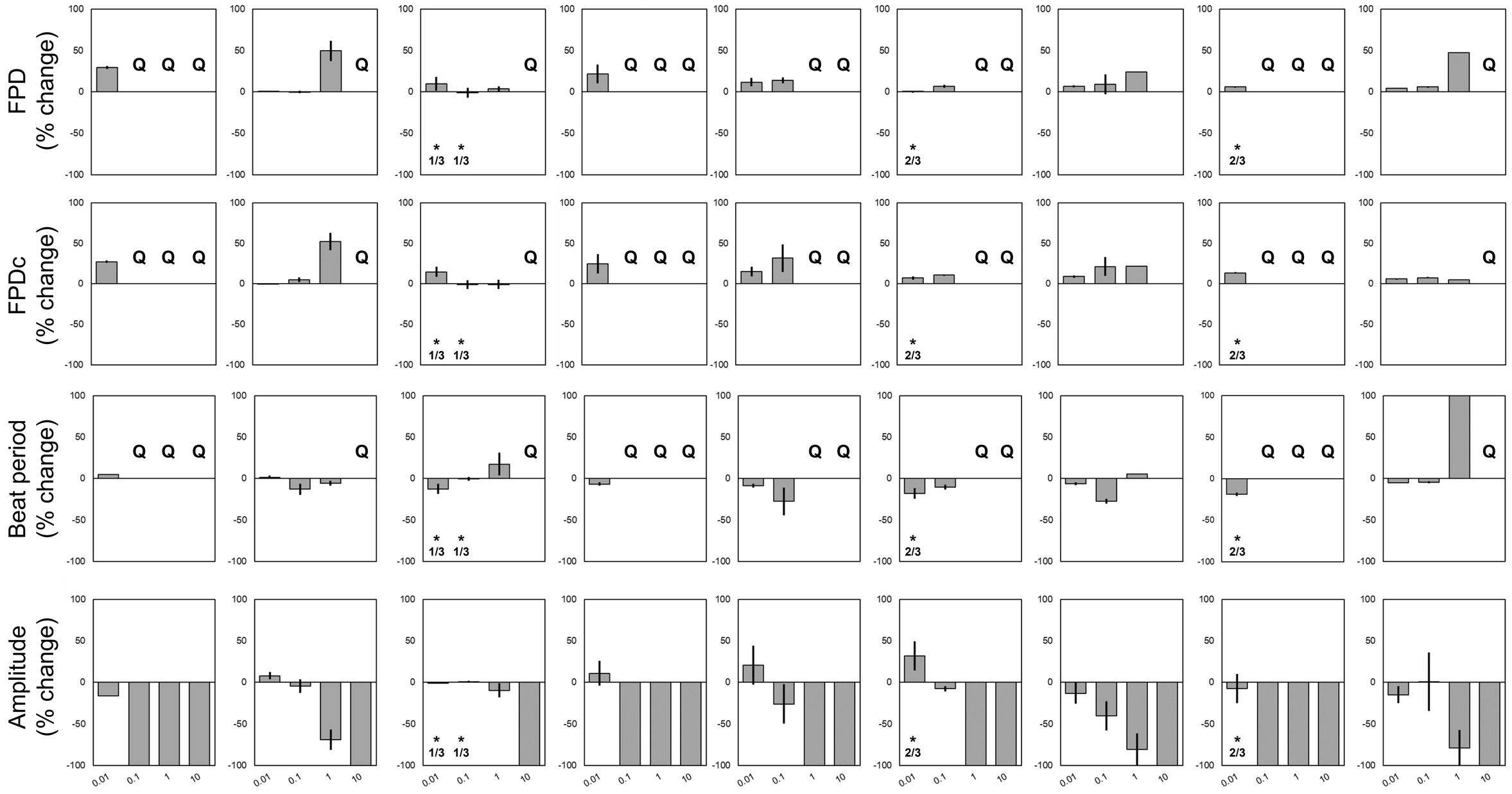
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Ca^{2+} channel blocker

Verapamil

Normal cell line | **LQTS cell line** | **BrS cell line**

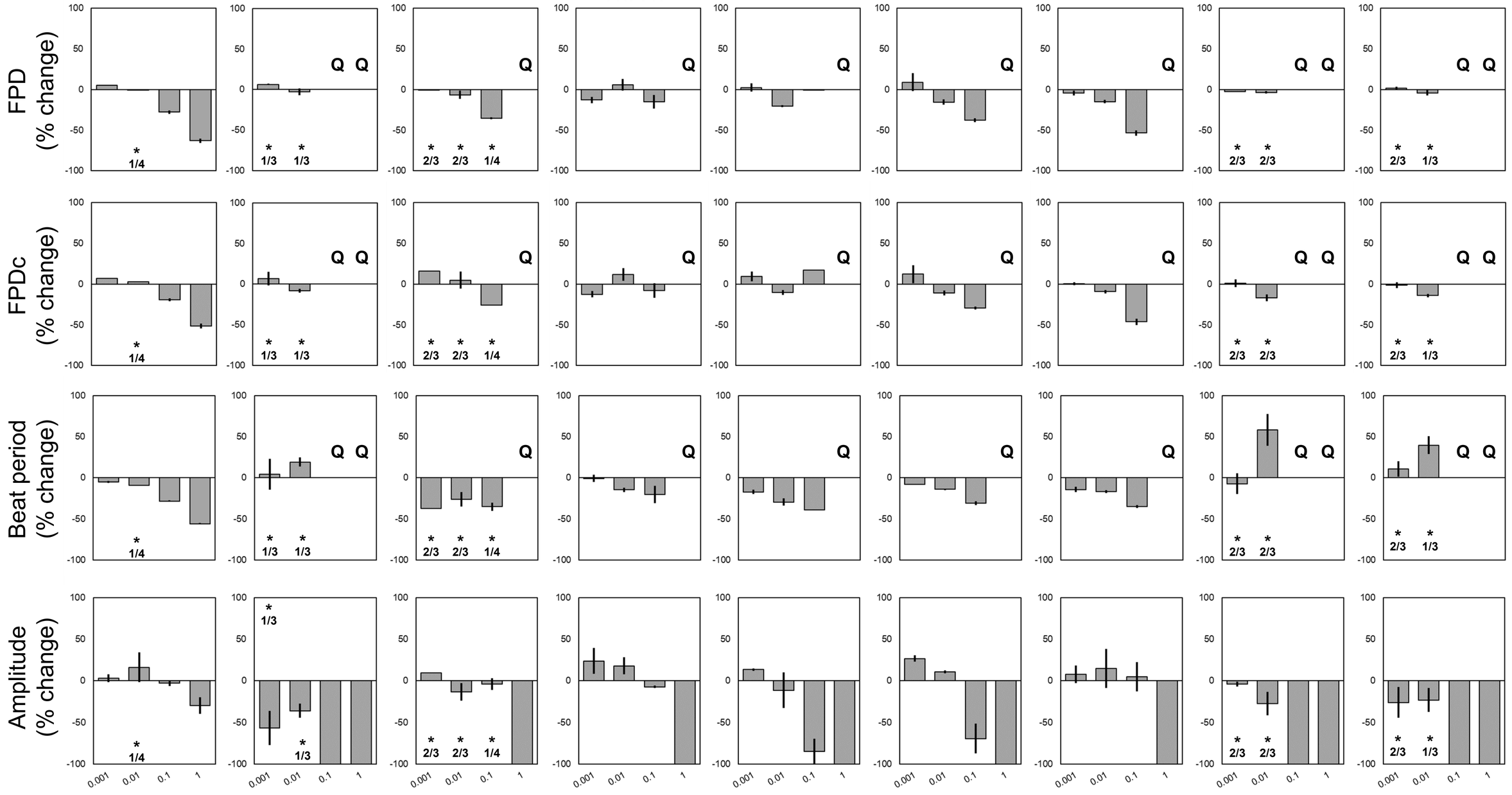
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Ca^{2+} channel blocker

Diltiazem

Normal cell line | **LQTS cell line** | **BrS cell line**

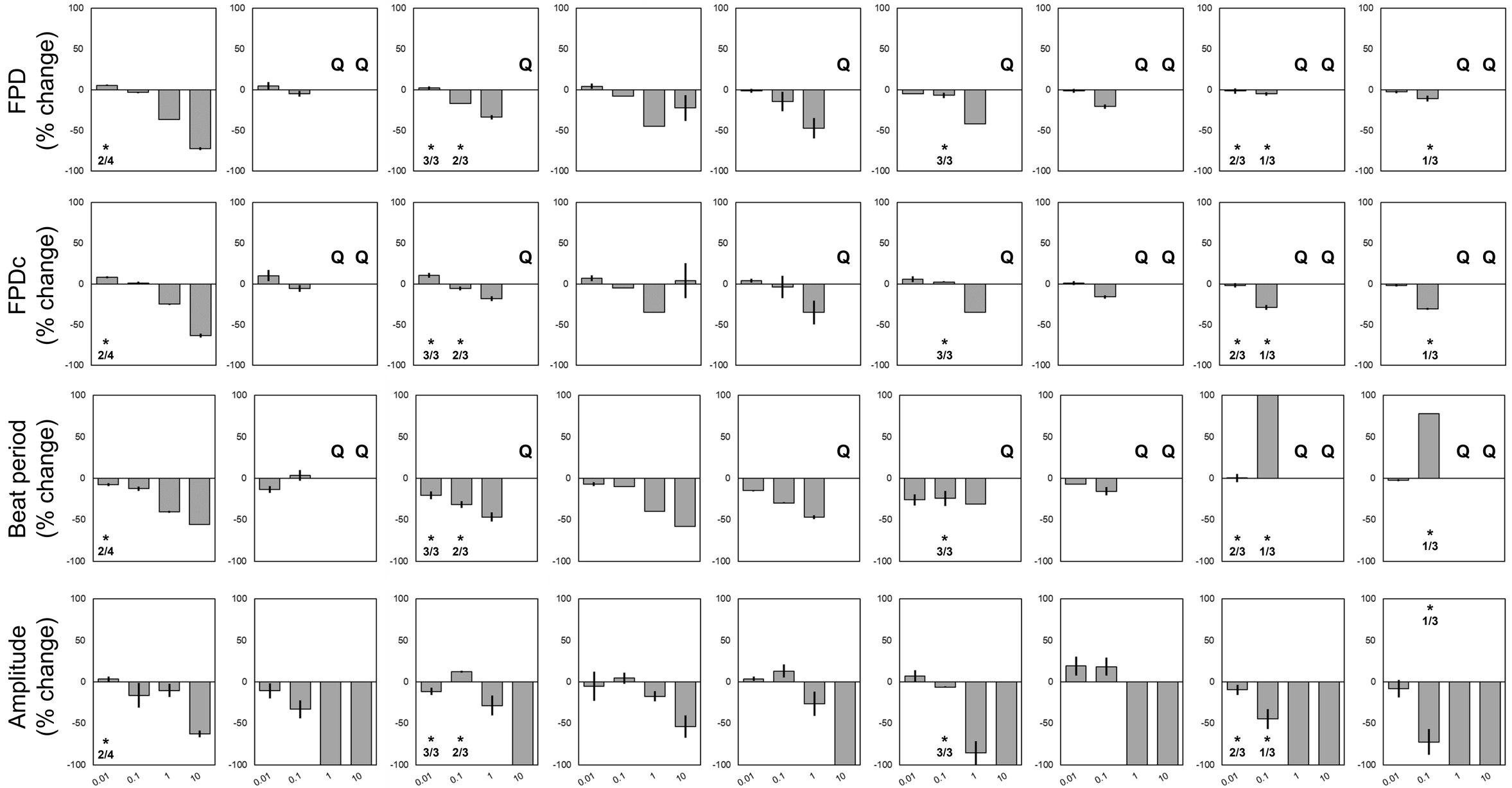
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Ca2+ channel blocker

Nifedipine

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

Q

* 1/3

* 1/4

* 3/3

0.001 0.01 0.1 1

Ca2+ channel blocker

Nitrendipine

Normal cell line

LQTS cell line

BrS cell line

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

* 1/3 | * 2/3 * 3/3 | * 2/3 * 1/3 | Q Q | Q | Q Q Q | Q | Q Q Q Q

0.01 0.03 0.1 0.3

Multi-channel blocker

Ibutilide

Normal cell line | **LQTS cell line** | **BrS cell line**

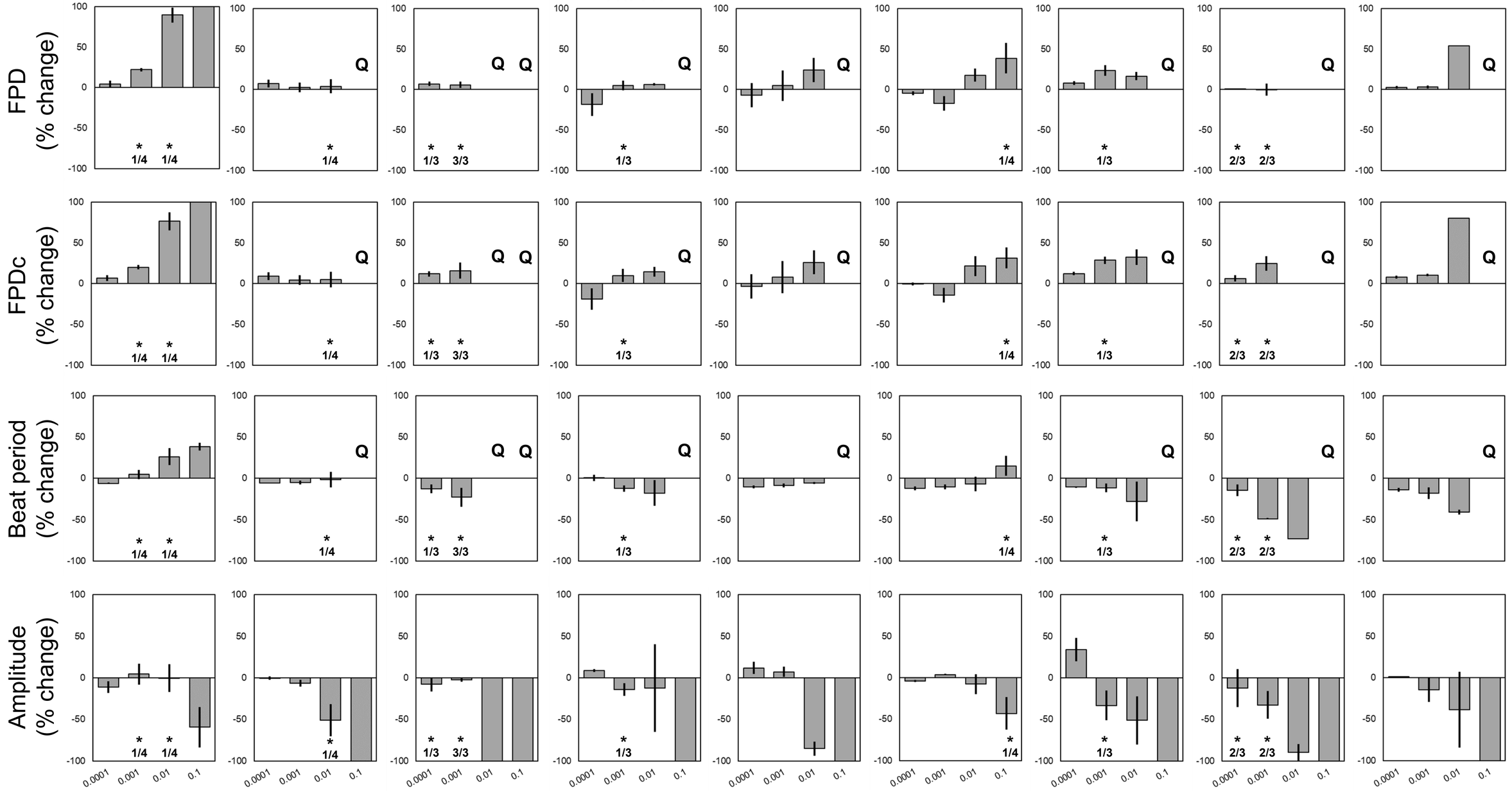
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Multi-channel blocker

Ranolazine

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

Q

* 1/3 * 3/3 * 1/3

* 1/3 * 2/3

* 1/4

* 1/3 * 2/3

* 2/3

0.1 1 10 100

Multi-channel blocker

Quinidine

Normal cell line

LQTS cell line

BrS cell line

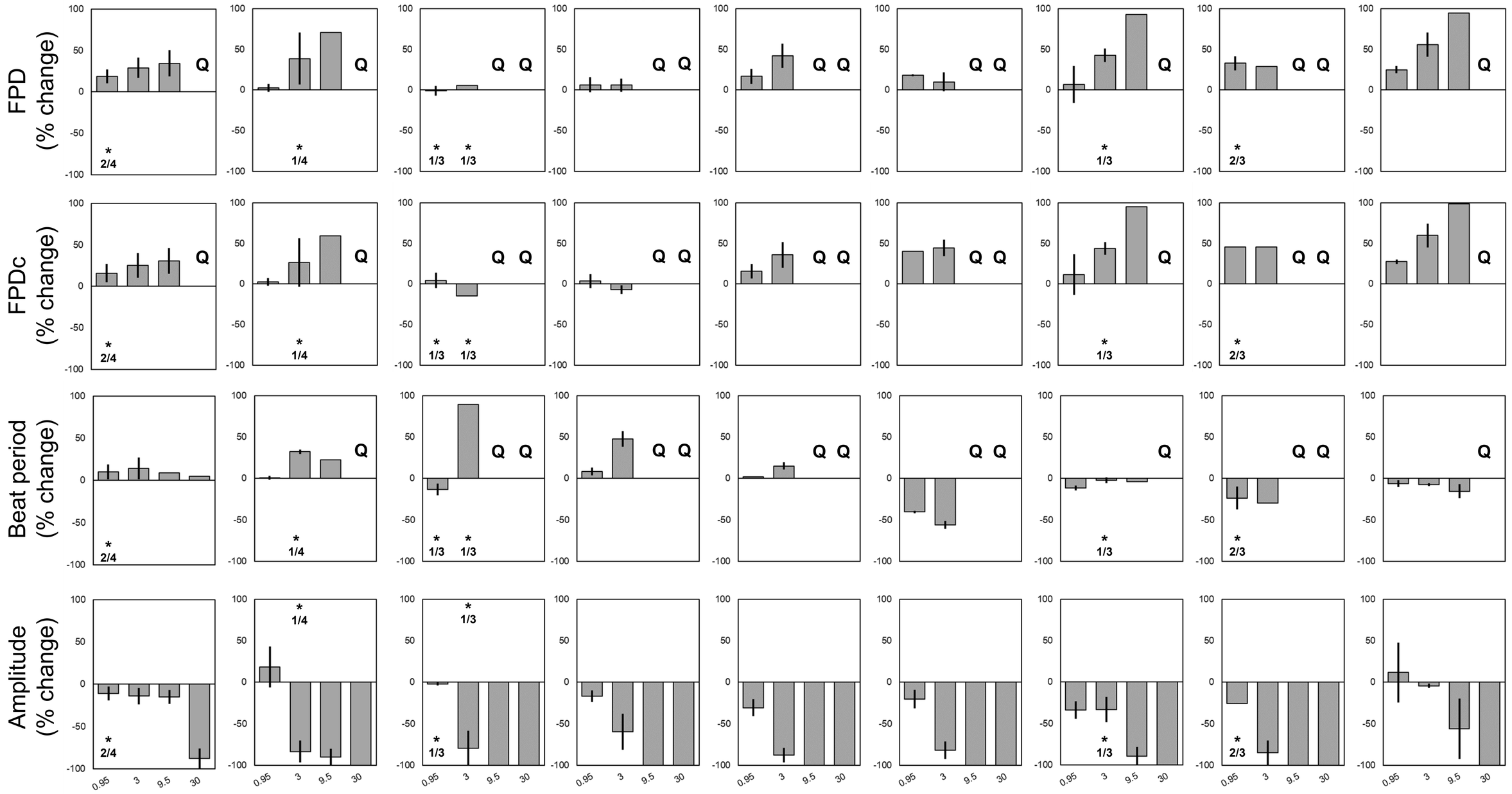
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Dopamine receptor

Domperidone

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

Q

* 1/4 * 1/3

* 1/3

* 2/4 * 3/4 * 1/5

* 1/3 * 1/3

* 1/3 * 3/3

0.003 0.03 0.3 3

Dopamine receptor

Droperidol

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

	iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
Q		1/4		1/3 3/3 1/3					
*	1/3		1/3 2/3			1/3 1/3 2/4	1/4	3/4 4/4	1/3 1/3

FPDc (% change)

	iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
Q		1/4		1/3 3/3 1/3					
*	1/3		1/3 2/3			1/3 1/3 2/4	1/4	3/4 4/4	1/3 1/3

Beat period (% change)

	iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
Q		1/4		1/3 3/3 1/3			1/4	3/4 4/4	
*	1/3		1/3 2/3			1/3 1/3 2/4			1/3 1/3

Amplitude (% change)

	iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
*	1/3	1/4	1/3 2/3	1/3 3/3 1/3		1/3 1/3 2/4	1/4	3/4 4/4	1/3 1/3

Each panel: 100, 50, 0, -50, -100; 0.03, 0.1, 0.3, 1

Dopamine receptor

Pimozide

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

0.001 0.003 0.01 0.03

* 1/3 * 1/4

* 1/4

* 3/3

* 2/3

* 2/3 * 3/4 * 1/4

Q

* 3/3

* 2/3

Dopamine receptor

Chlorpromazine

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 1/3	Q	* 1/3	Q	Q * 1/3	Q * 1/4	* 1/3 * 2/3 * 1/4	* 2/3	* 3/3 * 1/3 * 1/3

FPDc (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 1/3	Q	* 1/3	Q	Q * 1/3	Q * 1/4	* 1/3 * 2/3 * 1/4	* 2/3	* 3/3 * 1/3 * 1/3

Beat period (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 1/3	Q	* 1/3	Q	Q * 1/3	Q * 1/4	* 1/3 * 2/3 * 1/4	* 2/3	* 3/3 * 1/3 * 1/3

Amplitude (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 1/3		* 1/3		* 1/3	* 1/4	* 1/4 * 1/3 * 2/3	* 2/3	* 3/3 * 1/3 * 1/3

100 50 0 -50 -100

0.095 0.3 0.95 3

Dopamine receptor

Clozapine

Normal cell line | **LQTS cell line** | **BrS cell line**

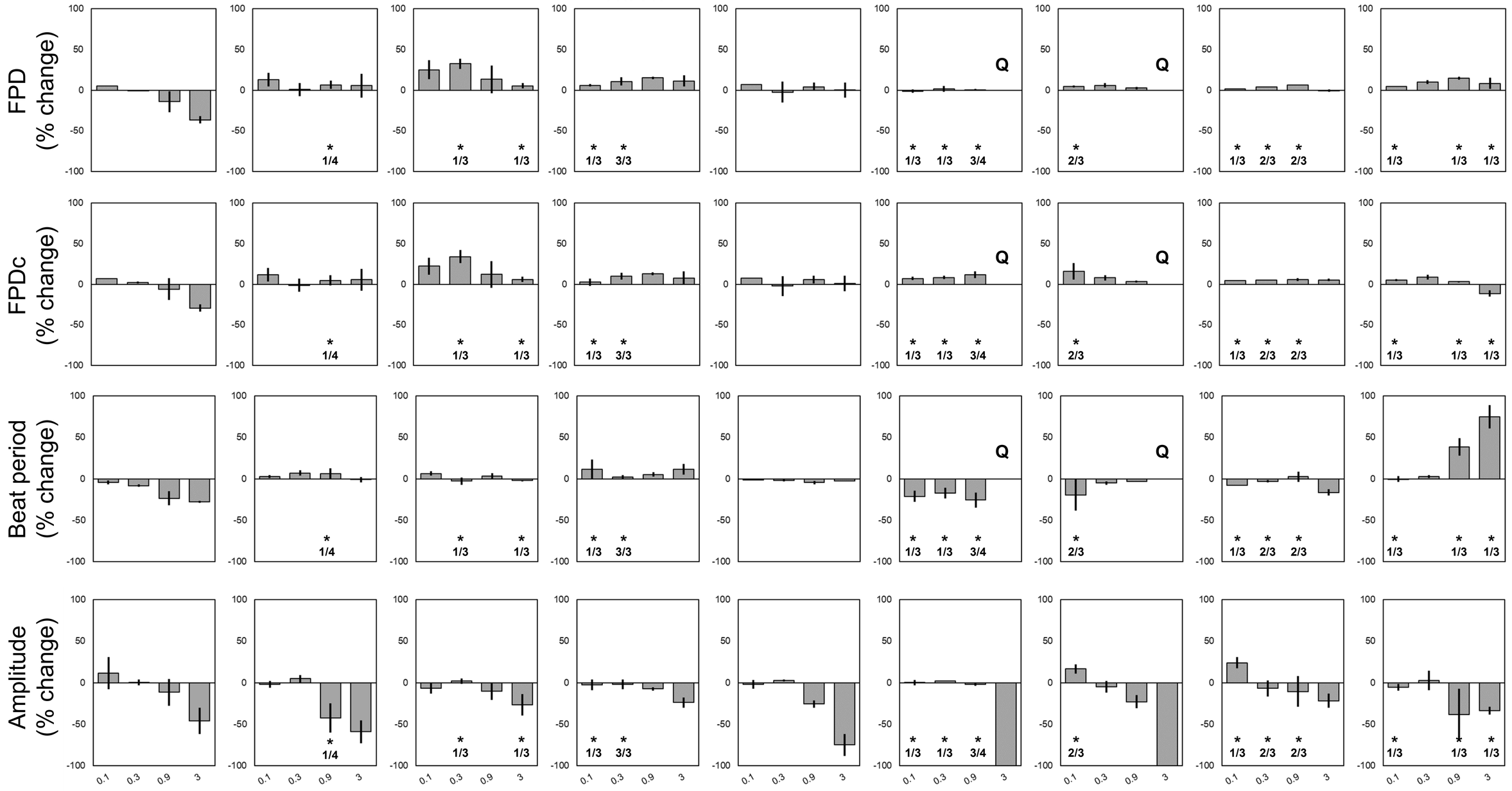
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Histamine receptor

Astemizole

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

0.0001 0.001 0.01 0.1

Q

* 1/3 * 2/3

* 1/4

* 2/3 * 1/3 * 1/3

* 2/4

* 3/3 * 2/3

* 1/3 * 1/3

Histamine receptor

Terfenadine

Normal cell line | **LQTS cell line** | **BrS cell line**

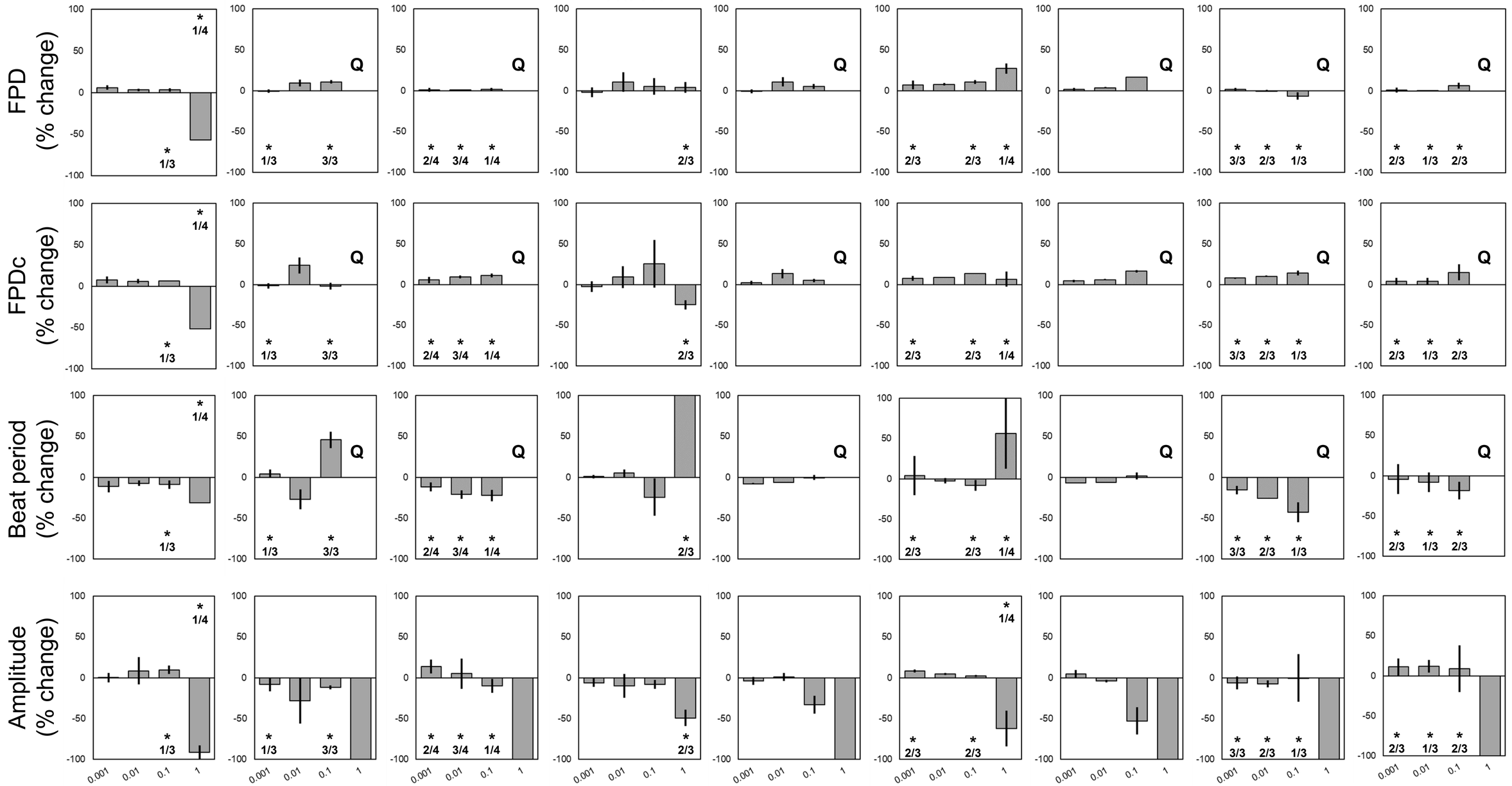
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Histamine receptor

Loratadine

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 2/4		* 3/4	* 1/3		* 3/3	* 1/3	* 2/3	* 1/3
* 2/4		* 3/4			* 3/3		* 2/3	* 1/3
* 1/4		* 3/4			* 2/3		* 3/4	* 1/3
* 3/5		* 2/4			* 3/3		* 2/4	

FPDc (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 2/4		* 3/4	* 1/3		* 3/3	* 1/3	* 2/3	* 1/3
* 2/4		* 3/4			* 3/3		* 2/3	* 1/3
* 1/4		* 3/4			* 2/3		* 3/4	* 1/3
* 3/5		* 2/4			* 3/3		* 2/4	

Beat period (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 2/4		* 3/4	* 1/3		* 3/3	* 1/3	* 2/3	* 1/3
* 2/4		* 3/4			* 3/3		* 2/3	* 1/3
* 1/4		* 3/4			* 2/3		* 3/4	* 1/3
* 3/5		* 2/4			* 3/3		* 2/4	

Amplitude (% change)

iCell	CMC11	PB01	TnR4F	DPHCi01	DPHCi08	DPHKi20	DPHCi15	DPHKi09
* 2/4		* 3/4	* 1/3		* 3/3	* 1/3	* 2/3	* 1/3
* 2/4		* 3/4			* 3/3		* 2/3	* 1/3
* 1/4		* 3/4			* 2/3		* 3/4	* 1/3
* 3/5		* 2/4			* 3/3		* 2/4	

0.001 0.003 0.01 0.03

Serotonin receptor

Ondansetron

Normal cell line | **LQTS cell line** | **BrS cell line**

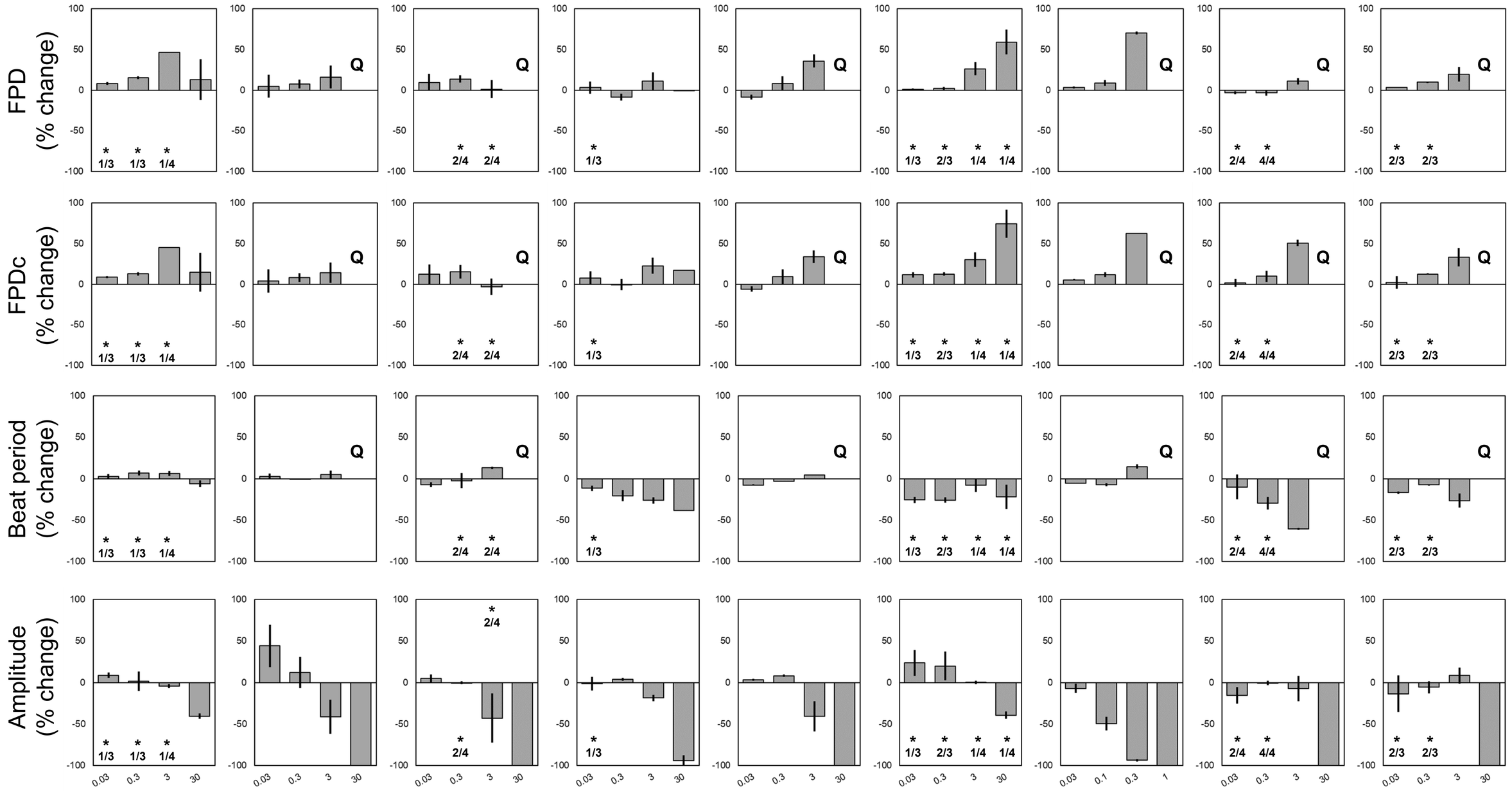
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Serotonin receptor

Cisapride

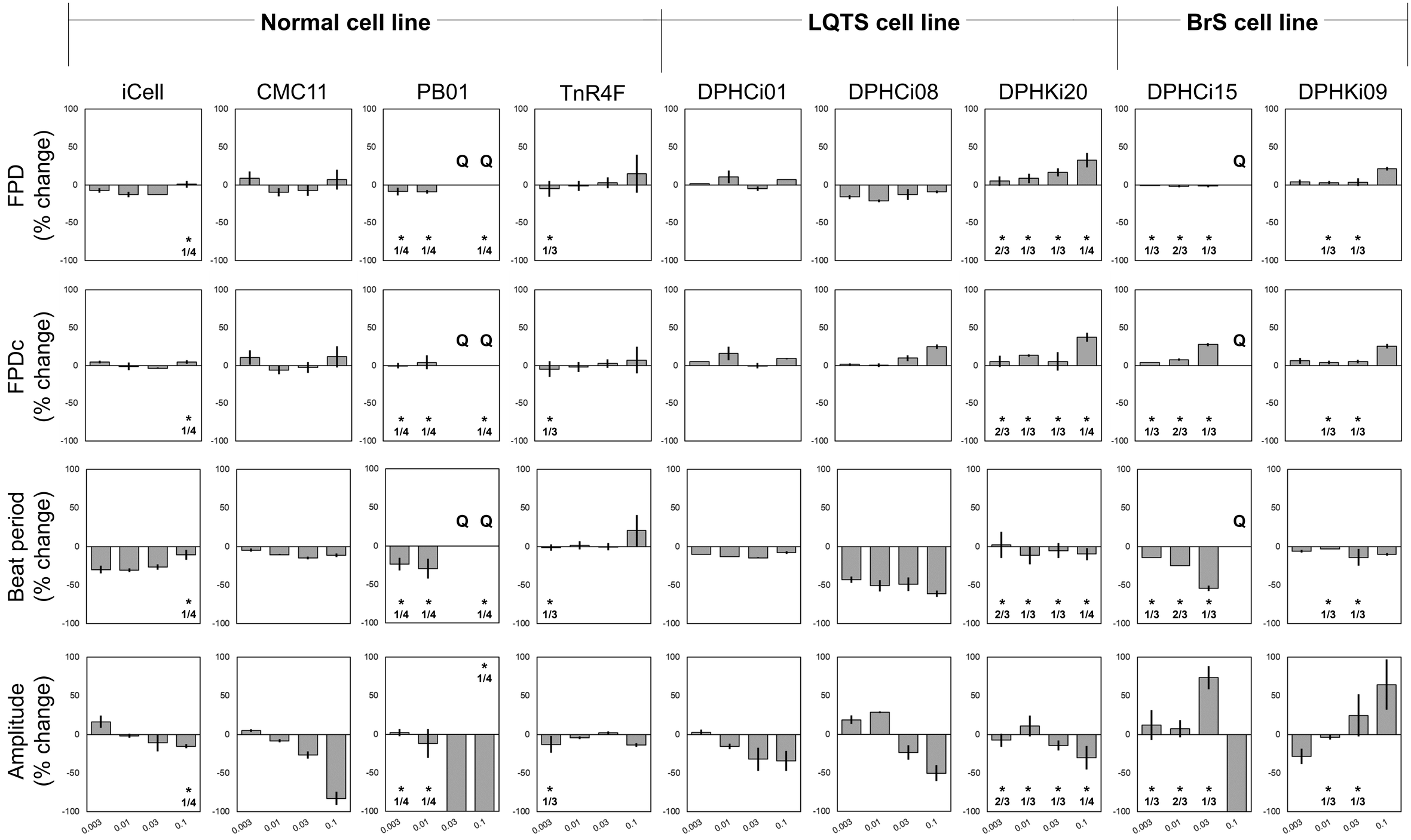
Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

Beat period (% change)

Amplitude (% change)



Serotonin receptor

Risperidone

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

* 1/3 * 1/3 * 1/3

* 1/3 * 3/4 * 1/3

* 1/3 * 2/3 * 1/3

* 1/3 * 1/3 * 2/4 * 2/4

* 2/3 * 1/3 * 2/3

* 3/3 * 2/4

0.003 0.01 0.03 0.1

Others

Vandetanib

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

Q

* 2/4

* 3/3 * 2/3 * 2/4 * 2/4

* 2/3 * 1/3 * 2/4

* 2/4 * 1/4

* 1/3

0.01 0.1 1 10

Others

Clarithromycin

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

100 50 0 -50 -100

0.1 1 10 100

Q

* 1/4 1/4

* 3/4 1/4 2/4 2/3

* 2/3 2/4

* 1/3 3/3 3/3

Others

Tamoxifen

Normal cell line | **LQTS cell line** | **BrS cell line**

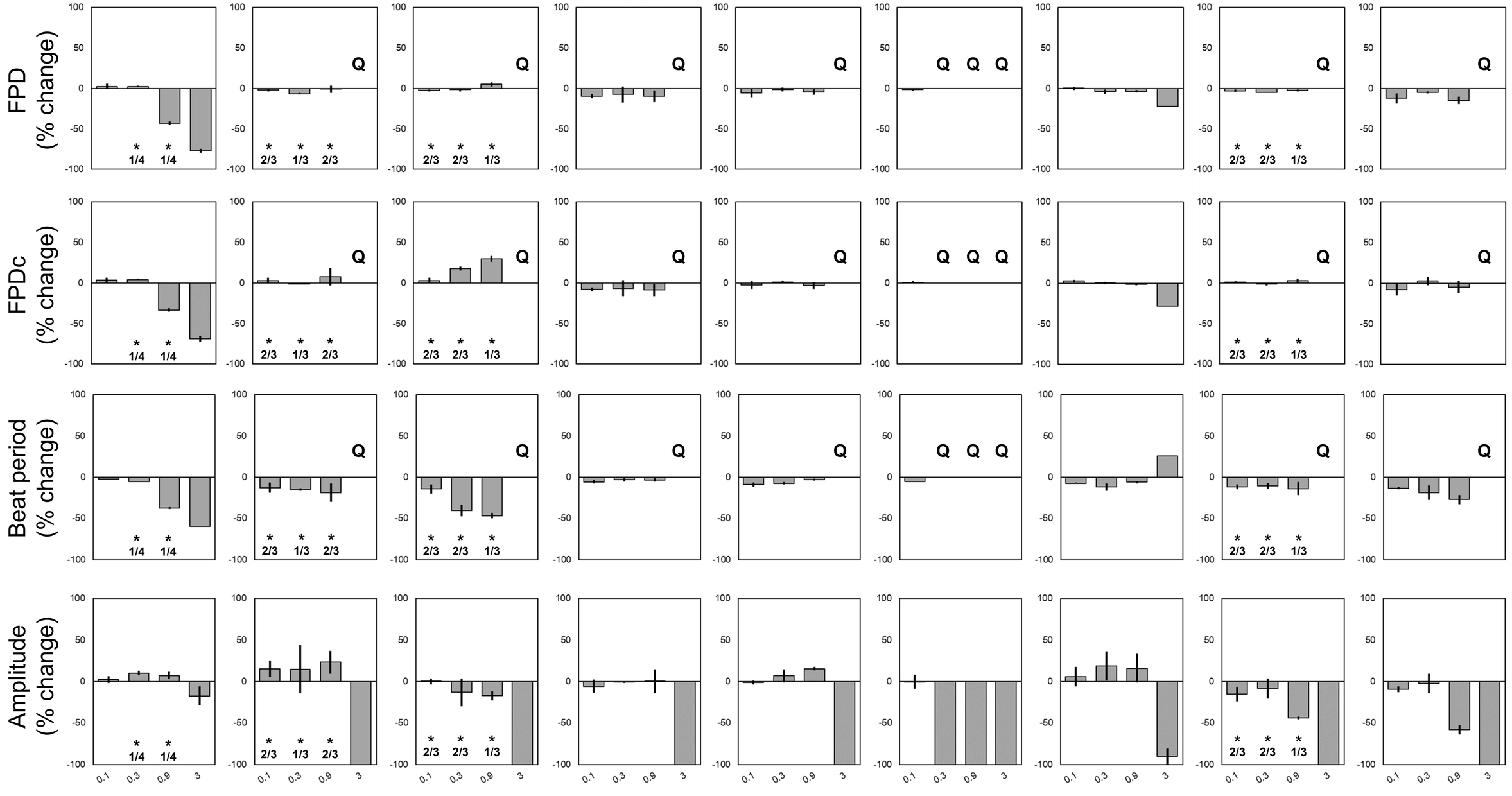
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



K+ channel blocker

Amiodarone

Normal cell line

LQTS cell line

BrS cell line

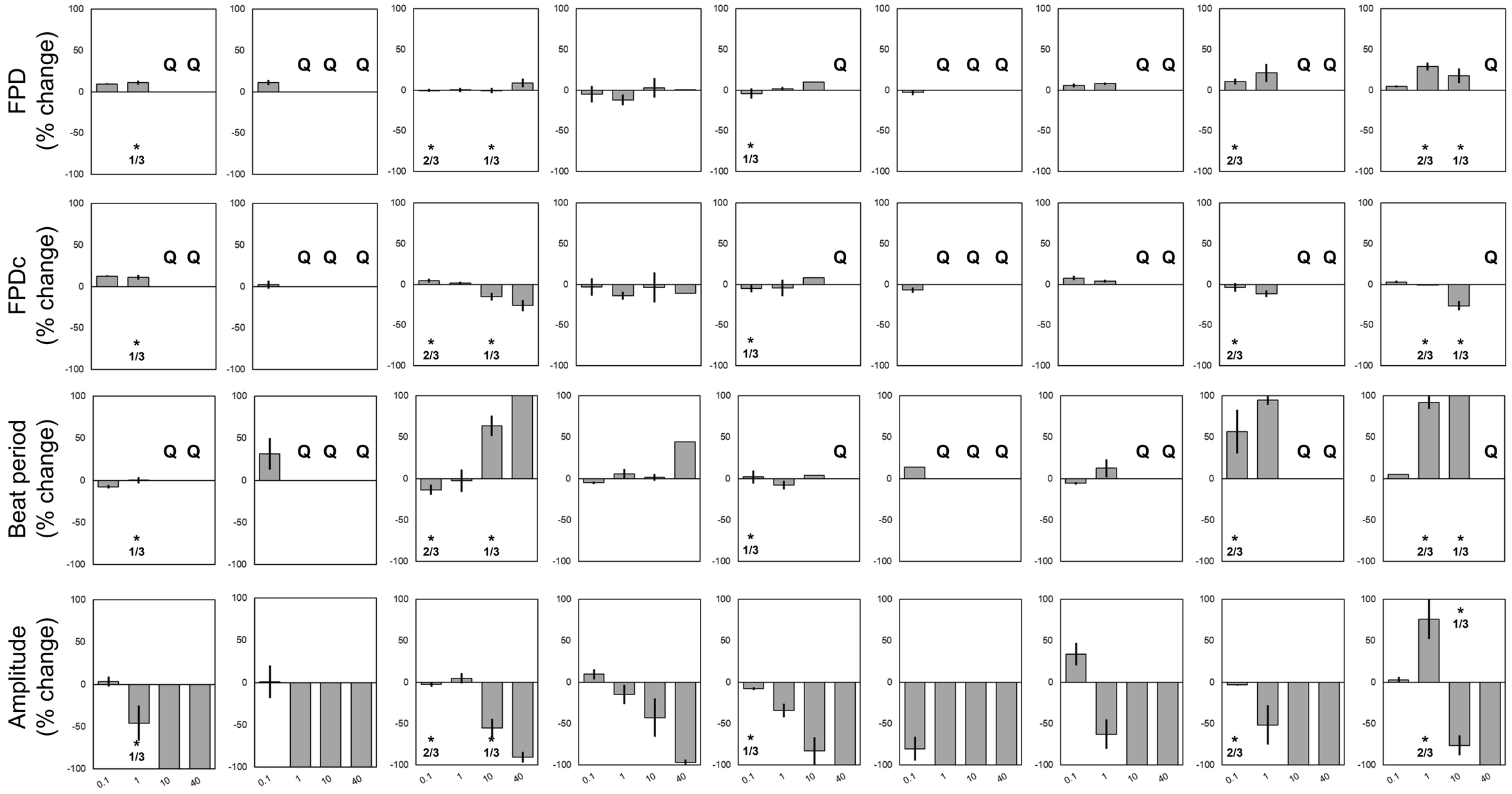
iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)



Antiviral agent

Remdesivir

Normal cell line | **LQTS cell line** | **BrS cell line**

iCell | CMC11 | PB01 | TnR4F | DPHCi01 | DPHCi08 | DPHKi20 | DPHCi15 | DPHKi09

FPD (% change)

FPDc (% change)

Beat period (% change)

Amplitude (% change)

